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Arithmetic word problems describing discrete quantities: E.E.G evidence for the construction of a situation model☆

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ABSTRACT

In this research, university students were asked to solve arithmetic word problems constructed either with discrete quantities, such as apples or marbles, or continuous quantities such as meters of rope or grams of sand. An analysis of their brain activity showed different alpha levels between the two types of problems with, in particular, a lower alpha power in the parieto-occipital area for problems describing discrete quantities. This suggests that processing discrete quantities during problem solving prompts more mental imagery than processing continuous quantities. These results are difficult to reconcile with the schema theory, according to which arithmetic problem solving depends on the activation of ready-made mental frames stored in long-term memory and triggered by the mathematical expression used in the texts. Within the schema framework, the nature of the objects described in the text should be quickly abstracted during problem solving because it cannot impact the semantic structure of the problem. On the contrary, our results support the situation model theory, which places greater emphasis on the problem context in order to account for individuals' behaviour. On a more methodological point of view, this study constitutes the first attempt to infer the characteristics of individual's mental representations of arithmetic text problems from EEG recordings. This opens the door for the application of brain activity measures in the field of arithmetic word problem.

1. Introduction

As testified by several international surveys, arithmetic word problem solving is an area of mathematics in which student performance is particularly low (e.g., Mullis, Martin, Foy, & Arora, 2012; OCDE, 2014). One of the main explanations of this poor performance is the difficulty that children encounter in the construction of an adequate mental representation from the text of the problem (Coquin-Viennot & Moreau, 2003; De Corte & Verschaffel, 1985; De Corte, Verschaffel, & De Win, 1985; Thevenot, 2008; Thevenot, Devidal, Barrouillet, & Fayol, 2007; Verschaffel, Greer, & De Corte, 2000). Therefore, examination of the relevance of different theories about the nature of the internal representation derived from a text problem is crucial to better understand and overcome children's difficulties.

Two main theories have been developed in order to account for arithmetic word problem solving. On the one hand, the schema theory states that word problems are solved through the activation of abstract frames from long-term memory (Kintsch & Greeno, 1985; Riley & Greeno, 1988; Riley, Greeno, & Heller, 1983). Those frames, or schemas, are triggered by the relational words used in the text of the problem, such as the mathematical expressions “more than”, “less than” or “altogether”. Once a schema is activated, the variables described in the text are entered into the empty slots of the frame until the problem is completely specified. At this point, the arithmetical procedure linked to the schema can be initiated and the calculations can be performed. On the other hand, the situation model theory hypothesizes that a non-mathematical representation is constructed before a more formal mathematical representation (Brissiaud & Sander, 2010; Nathan, Kintsch, & Young, 1992; Reusser, 1989; Staub & Reusser, 1995; Thevenot, Barrouillet, & Fayol, 2004; Thevenot & Oakhill, 2005). According to this theory, only the pieces of knowledge that are relevant for the understanding of the situation described in the problem are mobilized from long-term memory. The situation model is therefore mainly constructed ad-hoc in working-memory. Within this framework, the representation is constructed each time a problem is encountered, and its organization depends on specific contextual elements, which, in

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contrast, play very little role within the schema theory.

Evidence that the schema theory cannot always account for individuals' behaviour has been provided by Thevenot (2010) who demonstrated that, contrary to Kintsch and Greeno's assumption (1985), the mathematical expression used in a text of a problem does not play a key role in the solution process. In the experiment, the problem solving phase was followed by an unexpected recognition task of the problem questions. The results showed that the solvers were often confused when the mathematical expression used in the problem presented to them differed from the expression used in the original problem but depicted the same relation. For example, when the original problem was associated with the question "How many marbles do John and Paul have more than Bill?", participants often erroneously thought that the problem with the question "How many marbles does Bill have fewer than John and Paul?" was initially presented. This finding supports the view that only the relation between the elements described in a problem is mentally represented by individuals and not the specific mathematical expression used in the text. Again, these results are at odds with the schema theory, which considers the mathematical expression to be at the heart of the mental representation. On the contrary, they fit perfectly in a situation model framework according to which problem solvers construct an integrated mental representation of the situation described in the text (Bransford, Barclay, & Franks, 1972; Yuill & Oakhill, 1991). Within this framework, it is easily understandable that solvers are more likely to remember the gist of the problem than its literal formulation (Mani & Johnson-Laird, 1982; Zwaan & Radvansky, 1998).

In the current paper, we aim to test further the relevance of the schema and the situation model theories within the domain of arithmetic word problems. To this purpose, we manipulated the nature of the objects used in the text and reasoned that, if a situation model is constructed by individuals, it should influence the solution process. More precisely, we expect that continuous quantities (e.g., kilos of sand) and discrete quantities (e.g., number of apples) will not be mentally represented by individuals in the same way. Discrete quantities are countable and easily imageable (e.g., 3 apples) but continuous quantities are not associated with such precise representations and are therefore necessarily more difficult to visualize (e.g., 3 kilos of sand). Nevertheless, the schema theory would not predict differences in processing depending on the nature of the object described in the text because it is not supposed to affect the type of schema that will be triggered, which mainly depends on the mathematical expression used in the text. Therefore, the nature of the objects associated with quantities should quickly be abstracted during reading because it is not relevant for the solution task.

The level of mental imagery that word problems elicit seems difficult to investigate through behavioural studies. However, recording brain activity in adults during problem solving could be especially appropriate in order to examine our hypotheses. Therefore, we studied the level of mental imagery prompted by the nature of the objects described in the problem texts using electrophysiological responses. This is possible through the observation of different types of neural oscillations and more particularly of the alpha rhythm. The alpha band with a frequency going from 8 to 13 Hz was first thought to reflect cortical idling because it is recordable during the resting state (Pfurtscheller, Stancák Jr., & Neuper, 1996). Nowadays, electro-activity within the alpha band is more precisely considered as reflecting inhibition of the brain area in which it is recorded (Jensen & Mazaheri, 2010; Laufs et al., 2003; Toscani, Marzi, Righi, Viggiano, & Baldassi, 2010). Therefore and stated differently, when alpha activity is low, it is considered to reflect individuals' engagement in a cognitive activity associated with the recorded brain area. Accordingly, a decrease of the alpha power has been recorded when individuals engage in mental imagery (Barrett & Ehrlichman, 1982; Bértolo, 2005; Gualberto Cremades, 2002), especially in the parieto-occipital region, which is particularly involved in visual mental imagery (D'Esposito et al., 1997; Knauff, Kassubek, Mulack, & Greenlee, 2000; Kosslyn et al., 1999; Salenius, Kajola, Thompson, Kosslyn, & Hari, 1995; Short, 1953; Williamson, Kaufman, Lu, Wang, & Karron, 1997).

Consequently, if a situation model is constructed during problem solving, the level of imagery, and therefore the level of alpha activity, will vary with the nature of the objects described in the text of the problem. However, if problem solving consists in a more mechanistic activation of a schema from long-term memory, such a difference is not expected. In order to test these predictions, we asked adult participants to solve arithmetic word problems constructed either with discrete (e.g. apples) or continuous quantities (e.g. kilos of sand). Each problem describing discrete quantities was matched with a problem describing continuous quantities. Other variables, such as the number of words in the problems, the number of syllables and the frequency of the nouns associated with quantities were kept constant.

2. Method

2.1. Participants

Eighteen undergraduate students from the University of Geneva took part in the experiment for course credits. They were all native French speakers and they were aged between 18 and 29 (mean: 19.78 years, standard deviation: 2.49 years), three of them were men and three of them were left-handed.

2.2. Materials and procedure

Participants were asked to solve Compare problems involving static situations in which the unknown quantity could correspond to the difference between two sets (e.g., Joe has 6 apples. Tom has 4 apples. How many apples does Joe have more than Tom?), the compared set (e.g., Joe has 6 apples. Tom has 2 more apples than Joe. How many apples does Tom have?), or the reference set (e.g., Joe has 6 apples. He has 2 more apples than Tom. How many apples does Tom have?) (Riley et al., 1983). Twelve problems were constructed using discrete quantities such as *peaches* or *cars* and twelve problems were constructed using continuous quantities such as *kilos of sand* or *meters of rope*. Problems with discrete and continuous quantities were matched on the numbers of words contained in the text. Moreover, for a given problem, the nouns corresponding to the objects described in the text were matched on the number of letters, the number of syllables, the number of orthographic neighbours and had similar frequency ratio with a minimum of 28 occurrences per million (based on Frantext Corpus used in Lexique 2; New, Pallier, Brysbaert, & Ferrand, 2004). Moreover, the level of familiarity of each expression was assessed by 140 undergraduate students and an item-based analysis showed that it did not differ depending on quantity types (i.e, discrete or continuous), $F < 1$. Finally, the numbers used in corresponding problems (i.e. continuous vs. discrete) were the same (see Table 1 for an example). They comprised the numbers between 1 and 9, and the answer to the problems, which was reached either after a simple addition or subtraction, did not exceed 10.

Each problem was presented 4 times with different numbers across 4 experimental blocks. Therefore, each participant solved 96 problems: 4 (blocks) × 12 (problems) × 2 (quantity type: discrete or continuous). The order of presentation of the 24 problems (i.e., 12 with discrete and 12 with continuous quantities) was completely randomized in each block and a different randomization was created for each participant. Before the experiment, participants were presented with four warm-up trials.

The experiment was presented using E-Prime software. After a fixation cross lasting 500 ms, the text of the problem without the question was displayed on the screen. The question appeared after the participant pressed the enter key. As we will see in the next paragraph, this segmented presentation of the text allowed the isolation of a

Table 1
Example of a problem from the original experiment and its translation in English.

	Problem with a discrete quantity	Problem with a continuous quantity
Original language	La cousine de Joe achète 5 verres de lait. Elle achète 3 verres de lait de moins que la cousine de Tom. Combien de verres de lait la cousine de Tom achète-t-elle?	La demeure de Joe mesure 5 m de haut. Elle mesure 3 m de haut de moins que la demeure de Tom. Combien de mètres de haut la demeure de Tom mesure-t-elle?
English translation	Joe's cousin buys 5 glasses of milk. She buys 3 glasses of milk fewer than Tom's cousin. How many glasses of milk does Tom's cousin have?	Joe's house is 5 m in height. It measures 3 m less than Tom's house. How many meters in height is Tom's house?

precise and short period of time for the EEG recording. The answer to the problem, which always corresponded to a single digit number, was entered on the numerical keyboard by the participant. After the numerical key was pressed, a 1000 ms white screen was presented until the presentation of the next fixation cross.

Because we recorded EEG activity, the experiment took place in a dimly lit electromagnetically shielded room. We recorded the EEG signal using a system ActiveTwo from Biosemi with 32 electrodes. Four external electrodes were placed approximately 1 cm from the canthi of the eyes and under and below the left eye. These electrodes enabled the monitoring of eye blink and saccade movements. Two additional external electrodes were placed on the earlobes in order to set an external online reference during recording. The analysis performed on the alpha activity was restricted from the onset of the question until 1000 ms later. The rationale for choosing this precise recording period was that, for two thirds of the problems under study, the question could already be predicted from the text (either "Joe has 6 apples. Tom has 2 more apples than Joe" or "Joe has 6 apples. He has 2 more apples than Tom"). For the other third of the problems, the question was always the same (How many X have more marbles than Y?) and could therefore be predicted from the text just after a few presentations. Therefore, in the case of the mobilization of a schema for problem solving and at least for the vast majority of problems, the abstraction process should already have taken place before the question was read. More generally, we were interested in inferring the level of mental imagery from alpha bands once the mental representation of the problem had been constructed by participants to solve the problem and not while the mental representation was constructed. This is the reason why was recorded after the reading of the main text of the problem. Yet, recordings are of interest only if the mental representation is still active in working memory and it would therefore have been hazardous to measure brain activity after the problem had been solved. As a consequence, recordings of alpha band activity from the onset of the problem question seemed the most appropriate. The duration of one-second was chosen for the recording duration to avoid any contamination of brain activity by the presentation of the next problem.

The analysis was implemented in the BrainVision Analyzer 2 software (Brain Product GmbH; www.brainproducts.com). Too noisy electrodes were interpolated (min: 0, max 2 of the 32 electrodes) and then we used a mean reference. For the arithmetic task, we used a later filter of low cutoffs of 4 Hz and high cutoffs of 25 Hz with a slope of 48 dB/octave to exclude eyes or head movement artefact without losing information on the alpha rhythm and a notch filter at 50 Hz. We segmented the signal to epochs of 1000 ms., with respect to the conditions containing continuous or discrete quantities. We rejected segments that contained artefacts (i.e. jumps of > 100 μV), keeping a mean of 42.9 over 48 segments for discrete (89%) and of 43.8 segments for continuous quantities (91%) per participant. Finally, we applied Fourier transforms on each segment to quantify the frequency power spectrum. The analysis was restricted on the power corresponding to the alpha band (8 to 13 Hz).

Before the main task, each participant's resting state was recorded for 10 min. For 5 min, we asked participants to keep their eyes closed and for a further 5 min, they were asked to open their eyes while keeping quiet. A 2-minute break was allowed between the two recording parts. The analysis was similar to the one conducted during the main experiment except that we used a low cut-offs at 0.1 Hz and high cut-offs at 40 Hz, which was combined with an independent components analysis (ICA). We used such filtering because in the resting state, eye movements were almost absent. The segmentation of the resting state was also slightly different and consisted of epochs of 2-second segments with an overlap of 0.2 s between two adjacent intervals. Resting state activity was recorded because it allows independent measurement of maximal alpha activity. The difference in measurement between the condition with the eye opened and the eye closed is likely to exclusively reflect differences in visual processing (Ben-Simon et al., 2013). Therefore, electrodes for which this difference is the greatest are likely to record visual information and will be of special interest for the purpose of our study.

3. Results

An ANOVA was performed on the behaviorial data, with Quantity type (continuous or discrete) as a within-subject factor, on the percentages of correct responses in the arithmetic problem solving task. The analysis did not reveal a significant difference between the two conditions (96% and 95% for continuous and discrete quantities respectively), with $F < 1$. The same analysis was conducted on solution times, which were measured from the presentation of the text until the answer was given by participants, and the effect of Quantity type was not significant either (7506 ms and 7640 ms for the continuous and discrete quantity conditions respectively), $F(1, 17) = 1.72$, $p = .21$. However, an analysis of reading times of the problem text before the question appeared on screen revealed that participants took longer to read the text when it described discrete (5000 ms) rather than continuous quantities (4771 ms), $F(1, 17) = 12.38$, $p = .003$, $\eta^2p = 0.42$.

As explained in the Introduction, the EEG analyses were focused on the parieto-occipital area. More specifically, and as described in the Method section, electrodes of interest were selected depending on their visual processing involvement, which resulted in the selection of electrodes O1, OZ, O2, PO3, POZ, PO4, P7, P3, PZ, P4 et P8 (see the part circled in red in Fig. 1).

An ANOVA was conducted with Quantity type (continuous or discrete) as a within-subjects factor on the alpha power on the selected electrodes. Only the recordings associated with problems that were solved correctly were considered. A lower level of alpha activity was observed when problems describing discrete (0.78 μV^2) rather than continuous quantities (0.81 μV^2) were solved, $F(1, 17) = 4.55$, $p = .047$, $\eta^2p = 0.21$ (Fig. 2). This result was confirmed by a post-hoc non-parametric test of two dependent samples, $Z = 2.12$, $p = .03$.

As a follow-up analysis, we examined the possibility that our results were due to a difference in eye movements between the two conditions. Using an ANOVA with repeated measures, we extracted alpha power of the external ocular electrodes according to the Quantity type factor (i.e., discrete or continuous). Critically, there was no significant difference in the alpha power of the ocular electrode between the two conditions, $F < 1$. This result demonstrates that ocular movements are not at the origin of our effect.[1]

[1] In order to further examine the possibility of different alpha power depending on the conditions, an ANOVA also considering the side of the head

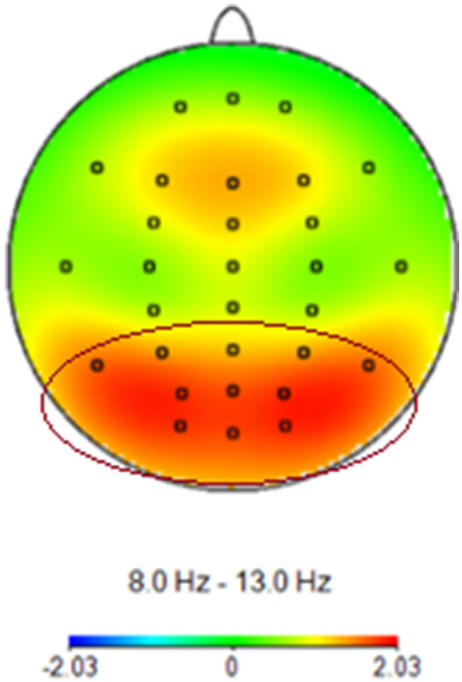

Fig. 1. Resulting alpha power during resting state after power in the condition eyes closed has been subtracted from power in the eyes open condition. The selected electrodes are circled in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

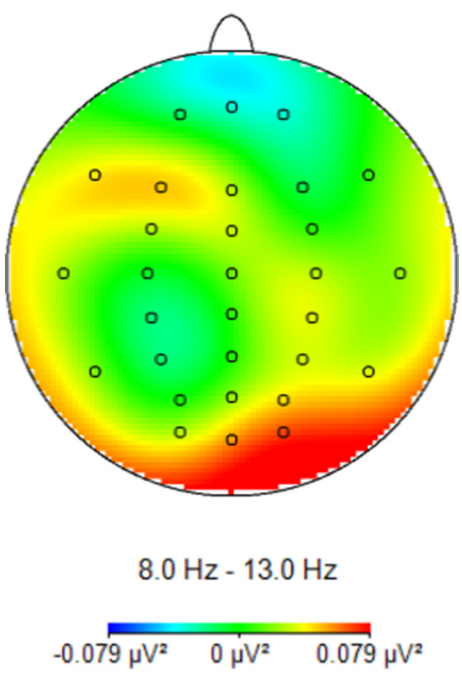

Fig. 2. Resulting alpha power for problems describing continuous quantities after subtraction of activation for problems describing discrete quantities.

4. Discussion

In this experiment, we examined the possibility that the mental representation constructed in order to solve arithmetic word problems might differ depending on the nature of the object described in the text. To explore this issue, university students were asked to solve a set of problems describing either continuous or discrete quantities and we measured the level of alpha power in the parieto-occipital area of their brain. This measure allowed a quantification of the mental imagery level during problem solving. As we expected, the level of alpha power was lower when discrete rather than continuous quantities were used in the texts, which, as explained in our Introduction, reflects a higher level of mental imagery. This result is coherent with the situation model framework, which hypothesizes that the context in which the mathematical situation is embedded influences the solving process. On the contrary, our results are difficult to reconcile with the schema theory, which cannot predict a difference in mental processing between problems describing continuous and discrete quantities. Indeed, according to this theory, empty slots in a fixed frame convoked from long-term memory are filled with quantities and objects described in the text of the problem as soon as they are encountered. Because the nature of the objects cannot have an impact on the problem solving process, they should be represented as abstract elements, or at the very least, they should rapidly lose their inherent characteristics during the course of the process. Thus, within the schema theory, since only the quantities and the procedures to be applied to those quantities (i.e., add or subtract them) are relevant for problem solving, the characteristics of the objects associated with the quantities and, specifically, their level of visualizability should not be mentally represented.

Therefore, the present study reinforces previous claims in the literature that the schema theory by itself cannot account for arithmetic word problem solving (e.g., Coquin-Viennot & Moreau, 2003; Staub & Reusser, 1995; Thevenot & Oakhill, 2006). Instead, an internal representation that is closer to the content of the problem seems to be constructed by individuals. Nevertheless, it should be noted that content effects have sometimes been interpreted within the schema framework. Indeed, Blessing and Ross (1996) argued that content effects could stem from the high correlation between the surface characteristics of the problems (e.g., type of objects) and their semantic structure. For example, problems in which a distance travelled has to be determined often involve trains. Hence, the nature of the objects used in the text could help individuals to correctly categorize the problem and mobilize the correct schema. However, in our experiment, only compare problems were used and, therefore, our participants had to solve almost 100 problems of the same semantic structure. Thus, within the context of this study, the categorization phase was of no relevance after a few problems had been solved. Therefore, we maintain our rationale that the schema theory leads to the assumption that the nature of the objects should have been neglected by our participants during the solving process. Still, our conclusion that the situation model theory better explains our results than the schema theory does not mean that schemata are never used when individuals solve arithmetic word problems. In fact, it has been shown that when the problems are or seem typical, individuals sometimes apply ready-made procedures, which can be triggered by problem schemata, without prior mental modelling of the situation (e.g., Lewis & Mayer, 1987; Verschaffel, De Corte, & Lasure, 1994). Such schemata activations can lead to errors due to misrepresentations of non-standard problems (see Jimėnez and Verschaffel (2014) for a literature review). Furthermore, we cannot exclude the possibility that, in our experiment, participants sometimes activated schemata from long-term memory in order to solve some of the problems. Still, our results show that situation models are, at least sometimes, constructed by individuals to solve familiar comparison problems.

A last point deserving discussion is related to the results we obtained at a behavioural level. Because we expected different processing

(footnote continued)
(left, right, center) as a within measure was conducted on our data The analysis shows an effect of quantity type, $F(1, 17) = 4.68$, $p = .045$, $\eta^2p = 0.22$, but no effect depending on the side of the brain, $F(2, 34) = 1.08$, $p = .35$ and no interaction between quantity type and side of the head, $F < 1$. Details on the level of the alpha rhythm can be found in the Appendix.

depending on the nature of the objects used in the text of the problems, we could have predicted differences in performance between problems involving continuous or discrete quantities. However, we chose to conduct our experiments in adults in order to ensure that they already had the possibility to extract abstract schemata from their numerous previous encounters with similar problems. The drawback in testing experts on simple problems, as attested by our results, is that they perform at ceiling levels and that variations in performance cannot really be expected. Yet, an analysis of reading times revealed that problems describing discrete quantities were read more slowly than problems describing continuous quantities. This result could reflect the time needed by participants to create an image of the discrete quantities. Interestingly, similar results were obtained by Knauff and Johnson-Laird (2002) who showed that reading the premises of a reasoning problem takes longer when the contents are easy to visualize (e.g., The dog is cleaner than the cat) rather than less visualisable (e.g., The dog is worse than the cat). Moreover, the authors showed that visualisable material slowed the processing of reasoning compared to less visualisable material. More precisely, inference problems such as "The dog is cleaner than the cat. The ape is dirtier than the cat. Does it follow: The dog is cleaner than the ape?" are solved more slowly than problems such as "The dog is better than the cat. The ape is worse than the cat. Does it follow: The dog is better than the ape?". Knauff and Johnson-Laird logically concluded that mental imagery impedes reasoning. According to the authors, the visual details contained in mental images are far less relevant for the reasoning process than more abstract mental representations, which can represent any possible situations. Therefore, our results could echo Hegarty and Kozhevnikov's results (1999) showing that imagery generated by sixth-graders is detrimental to their performance or, more generally, conclusions that providing children with pictorial representations does not always help them to solve the problems (e.g., Dewolf, Van Dooren, & Verschaffel, 2017; Van Lieshout & Xenidou-Dervou, 2018). Nevertheless, as noted by Van Garderen and Montague (2003), it might not be the visual image per se that is responsible for the decrease in performance but rather the fact that an image is generated instead of the correct structural representation. As a matter of fact, several studies have shown that eliciting accurate visual-schematic representations (or mathematical drawings) rather than pictorial representations (or situation drawings) facilitates problem comprehension (Boonen, Van Wesel, Jolles, & Van der Schoot, 2014; Vicente, Orrantia, & Verschaffel, 2008). Unfortunately, it is difficult to determine whether the manipulations we made in the present research impacted individuals' mental representations of the objects or of the relationships between the objects. In either case, our results could have positive implications from a pedagogical point of view and further studies in children using our paradigm, even at a simple behavioural level, could be informative about the precise role of mental imagery in word problem solving performance. Another way to answer the question about the impact of manipulating the nature of the quantities described in the text could be to expand our analyses to additional frequency bands. For example, examining theta activity could be informative because it is sometimes viewed as reflecting cognitive load (Jensen & Tesche, 2002; see Hinault & Lemaire, 2016 for a review about cognitive interpretations of rhythmic activities in numerical cognition). Another alternative approach in order to address our question would have been to conduct an event-related potential (ERP) analysis, which could have allowed us to contrast the EEG signal patterns for problems with continuous and discrete quantities. Nevertheless, ERP analysis is generally used for more homogeneous tasks (e.g., Burra & Kerzel, 2014 for a visual perception task or Hinault, Lemaire, & Phillips, 2016 for a computational estimation task) than arithmetic word problem solving tasks in which text reading comprehension and calculations are required. In fact, we have to acknowledge that studying such a complex cognitive process required a few comprises in order to maintain an acceptable ecological validity of the task while collecting meaningful EEG data. For example, we decided to present the text of the problem in two different segments, which is not usual, but allowed us to better identify the different processing steps in problem solving. In the same way, we decided to record brain activity during 1 sec at the beginning of the question presentation but this choice can be debated. Still, this pioneering study reveals that brain activity recording technics can be applied in addition to more classical behavioural methods in the domain of arithmetic word problem solving and can provide new theoretical knowledge within this field.

Compliance with ethical standards

The local ethics committee approved our study, which was therefore conducted in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki. All our participants gave their informed consent prior to their inclusion in the study.

Declaration of interest

None.

Appendix A. Alpha power in μV^2 by hemisphere and by types of quantities

	Hemisphere		
Type of quantities	Left	Center	Right
Discrete	0.83	0.75	0.81
Continuous	0.84	0.77	0.84

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